

Curated Research Articles

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- **“Lithium-Free” Anode Solid-State Batteries Enabled by 3D Structured MIEC Garnet** — score: 1.000 The study introduces a mixed ionic and electronic conducting (MIEC) garnet scaffold that enables stable lithium formation in “lithium-free” solid-state batteries. By facilitating uniform lithium nucleation within a 3D porous structure, the technology achieves exceptionally low anode impedance and remarkable Coulombic efficiencies, allowing for efficient cycling and highly reversible lithium plating without pressure at room temperature, which could significantly influence the design of future solid-state battery systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Fluorination-Tuned Covalent Organic Framework Interphases for Controllable LiF Content in Dendrite-Free Lithium Metal Batteries** — score: 1.000 This study introduces a fluorination-gradient approach using covalent organic frameworks (COFs) to create an artificial solid electrolyte interphase (SEI) in lithium metal batteries, enabling controlled LiF content. The designed interphase enhances ion transport and stability, significantly reducing dendrite formation and improving battery performance, demonstrated through extended cycling stability and capacity retention across various battery chemistries. This innovative strategy paves the way for safer and more efficient lithium metal batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Synergetic Regulation of Enthalpy-Entropy via Dipole Interactions Enabling Textured and Low-Temperature Aqueous Zinc-Ion Batteries** — score: 1.000 This article presents a hydrated eutectic high-entropy electrolyte that effectively utilizes dipole interactions to regulate enthalpy and entropy in aqueous zinc-ion batteries (AZIBs). By creating diverse solvation environments and achieving interfacial regulation, the electrolyte mitigates issues of uneven zinc deposition and parasitic reactions, enabling stable performance at both standard and sub-zero temperatures. The results indicate enhanced durability and efficiency of zinc plating, significantly improving the operational lifespan of these batteries in extreme conditions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Multi-Step Oxygen Redox Mechanism in the Polyanionic Lithium-Rich Cathode $\text{Li}_2\text{FeSiO}_4$** — score: 1.000 This study investigates the top-of-charge behavior of the lithium-rich silicate cathode $\text{Li}_2\text{FeSiO}_4$, revealing that despite its stable polyanionic framework, it can still undergo oxygen redox processes leading to spontaneous O_2 formation. Using density functional theory and ab initio molecular dynamics, the research uncovers a multi-step mechanism involving iron ion migration and redox reactions, challenging the assumption that lithium-rich silicates are immune to oxygen release during charging cycles. Journal: *Journal of the American Chemical Society ASAP*
- **Disentangling Catalytic Activity and Film Conductivity Reveals Temporally Decoupled Catalyst Evolution and Conductivity Artifacts** — score: 1.000 The article presents a novel technique called operando Fourier-transform electrochemical source–drain impedance spectroscopy (FT-ESDIS), which simultaneously measures the conductivity and charge-transfer resistance of electrocatalysts in real-time. The study shows that the evolution of catalytic activity and charge transport in nickel and iron hydroxides occurs on different time scales, challenging previous assumptions about their interdependence and revealing that apparent conductivity changes may primarily result from electrolyte-mediated artifacts rather than direct material conductivity changes. This approach enhances the understanding of electrocatalyst performance in energy conversion systems by allowing for more precise assessments of transport and reactivity. Journal: *Journal of the American Chemical Society ASAP*
- **Bulk/Surface Coengineering in Layered Oxides via Selective Niobium Doping to Enable Durable High-Energy Sodium-Ion Pouch Cells** — score: 1.000 The article discusses how selective niobium doping in sodium-ion battery cathodes enhances their performance by improving structural stability and reducing transition metal dissolution. This dual engineering strategy increases the P2 phase ratio and forms a protective layer, enabling durable operation at high voltages (4.3 V) and achieving stable cycling over 600 cycles, while ensuring safety during operation. Journal: *Wiley:*

- **Nano-Crosslinked Polyether-Based Gel Polymer Electrolytes for Interface-Stable High-Voltage Lithium-Ion Batteries** — score: 1.000 The article presents a novel nano-crosslinked gel polymer electrolyte created through a two-step process that enhances the mechanical strength and ionic conductivity crucial for high-voltage lithium-ion batteries. By employing LAP nanosheets to create a robust network and regulate lithium ion solvation, the electrolyte significantly reduces side reactions at the electrodes, yielding impressive cycling longevity and stability across multiple battery configurations. This approach not only improves performance but also provides a new framework for developing advanced gel polymer electrolytes in battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Electrolyte Interfacial Reactivity Regulates Alloying Anode Pulverization** — score: 1.000 The study investigates how the reactivity of electrolytes affects the performance of alloying anodes, specifically using bismuth in sodium-ion batteries. It finds that a low-reactivity electrolyte supports long-term cycling with minimal capacity loss by allowing porous structures and thin solid-electrolyte interphases (SEIs), while high-reactivity electrolytes lead to rapid capacity degradation due to dense packing and thick SEIs. This approach can inform the design of more stable, high-energy battery systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Decoupling Quasi-Solid-State Sulfur Conversion Mechanisms in Weakly Solvating and Locally High Concentration Electrolytes** — score: 1.000 This study compares two electrolyte formulations for lithium-sulfur batteries: a weakly solvating CME-TTE and a localized high-concentration G2-TTE. The findings reveal that G2-TTE enables superior lithium-sulfur chemistry through multi-dentate coordination and effective anion participation, significantly reducing polysulfide crossover and improving cycling performance, achieving capacities exceeding 900 mAh g⁻¹ after 200 cycles with high efficiency. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Ion Association Dynamics Governs Low-Temperature Ion Transport of Sodium-Ion Battery Electrolytes** — score: 1.000 The study investigates the mechanisms behind the decline in ion transport in sodium-ion battery electrolytes at low temperatures. Utilizing a combination of machine-learned molecular dynamics and Raman spectroscopy, the researchers find that while the coordination between sodium ions and solvent remains stable, increased ion association and prolonged lifetimes of these pairs at lower temperatures significantly hinder the mobility of Na⁺ ions, ultimately reducing electrolyte conductivity. This highlights the importance of ion-association dynamics over static solvation structures in understanding low-temperature performance. Journal: *ACS Energy Letters ASAP*
- **Na-Ion Cathodes in Minutes: Microwave Synthesis across the Na-Mn-Fe-Ni-Cu-O Composition Space** — score: 1.000 This study presents a microwave synthesis method for rapidly producing layered NaMO₂ cathode materials for Na-ion batteries, achieving results in just 7-10 minutes across various compositions. Characterization techniques confirm the successful formation of well-ordered frameworks without the need for excess sodium, while electrochemical tests highlight the influence of composition on Na-ion behavior, such as capacity and operating potential, showcasing the method's potential for efficient exploration of battery materials. Journal: *ACS Applied Energy Materials ASAP*
- **Demarcating the Effects of Aliphatic Ester Chain Lengths and Molecular Weight on Electrode-Electrolyte Interphase Formation for Wide-Temperature Operation of Lithium-Ion Batteries** — score: 1.000 This study investigates the impact of aliphatic ester chain lengths and molecular weights on the formation of the electrode-electrolyte interphase in lithium-ion batteries, aiming to enhance performance across a broad temperature range of -60 to +60 °C. The findings reveal that optimized chain configurations improve low-temperature conductivity and discharge capacities while highlighting the importance of reactivity in electrolyte design. Moreover, ternary solvent blends were developed to balance high-temperature cycling performance with low-temperature efficiency, laying the groundwork for designing more effective electrolytes. Journal: *Journal of the American Chemical Society ASAP*
- **Competitive Coordination Decouples Charge Transport from Viscous Relaxation in**

Water-in-Salt Electrolytes — score: 0.900 The study investigates how zwitterionic additives affect the properties of concentrated LiTFSI water-in-salt electrolytes, revealing that while these additives decrease absolute conductivity, they transition the electrolytes from subionic to superionic behavior. This occurs due to competitive coordination dynamics, where interactions between zwitterions and lithium ions impact mobility and viscosity, leading to distinct changes in solvation and ion transport characteristics. Journal: *Journal of the American Chemical Society ASAP*

- **Enhancing the Stability of Pt/Carbon with Ti₃C₂ MXene-Derived TiO₂ for the Oxygen Reduction Reaction in Acidic Media** — score: 0.900 This study explores enhancing the stability and activity of platinum catalysts supported by carbon through the incorporation of titanium carbide MXene-derived titanium dioxide (TiO₂). Characterization techniques revealed that this novel support structure minimizes platinum particle growth and supports improved durability under stress tests, outperforming traditional platinum on carbon catalysts in fuel cell applications. Journal: *Journal of the American Chemical Society ASAP*
- **Substrate-Enhanced Atomic Layer Deposition of Lithium Niobate on NMC Cathodes for Sulfide-Based All-Solid-State Batteries** — score: 0.900 The article explores the substrate-enhanced atomic layer deposition (ALD) of lithium niobate (LNO) on NMC cathodes to address the interface stability issues in sulfide-based all-solid-state batteries. By using commercial precursor chemistry, the researchers achieved high-quality ultrathin LNO coatings, which enhanced the discharge capacity of the batteries by up to 20% and improved their electrochemical performance by mitigating electrolyte decomposition during cycling. Journal: *ACS Energy Letters ASAP*
- **Escaping the Polymerization-Kinetics Trap in Ether Electrolytes for Durable Silicon Interphases** — score: 0.900 The study reveals that stronger polymerization in ether electrolytes does not guarantee better durability for silicon anodes, as demonstrated by the contradictory performance of trioxane compared to dioxolane. By adding lithium nitrate, which stabilizes the polymerization process while enabling effective passivation of the silicon interphase, the researchers achieved a more coherent interface and enhanced cycling stability, resulting in significant retention of capacity over multiple charge-discharge cycles. Journal: *ACS Energy Letters ASAP*
- **Molecular-Level Cleavage and Kinetics Optimization in PVDF Electrolytes for Anode-Lean Ultrastable Solid-State Sodium Batteries** — score: 0.900 The study presents a novel approach to enhance the performance of PVDF electrolytes in solid-state sodium batteries by incorporating Na₃P as a nanofiller. This dual-function filler cleaves C-F bonds, enhancing ionic conductivity by improving Na⁺ mobility and facilitating the formation of a stable solid-electrolyte interphase, leading to batteries with remarkable stability and longevity, demonstrating minimal capacity fade over extensive cycling. Journal: *Journal of the American Chemical Society ASAP*
- **Weak-Solvation Molecular Ion Gates Enabling Multipath Li⁺ Hopping and Air Stability in Garnet/Poly(ethylene oxide) Electrolytes** — score: 0.900 The study introduces a weak-solvation molecular ion gate (WSMI) approach to enhance Li⁺ transport in poly(ethylene oxide) (PEO)-based electrolytes by modifying Li_{6.4}La₃Zr_{1.4}Ta_{0.6}O₁₂ with 4-methoxy-2-(trifluoromethyl)benzoic acid. This strategy improves air stability and ion hopping by eliminating blocking layers and facilitating coordination exchanges, resulting in a high transference number of 0.622 and significant cycling stability in battery cells. Journal: *Journal of the American Chemical Society ASAP*
- **Revealing the Proton Slingshot Mechanism in Solid Acid Electrolytes through Machine-Learning Molecular Dynamics** — score: 0.900 The study uses machine-learning molecular dynamics to investigate the mechanisms behind fast proton conduction in solid acid electrolytes, specifically CsH₂PO₄ and CsHSO₄. It uncovers a “proton slingshot mechanism” where polyanion rotation facilitates proton transport, while revealing differences in transport dynamics linked to proton concentration and polyanion structure. The findings provide a comprehensive understanding of how polyanion dynamics and proton coordination influence conductivity, suggesting potential methods for enhancing performance through proton concentration adjustments. Journal: *Journal of the American Chemical Society ASAP*

- **Impact of Solvent Impurities from Upscaled Synthesis and Their Purification in Lithium Metal Batteries** — score: 0.900 The article examines the challenges of scaling the synthesis of fluorinated ether electrolytes for lithium metal batteries, specifically focusing on the impact of solvent impurities. Using the model solvent F5DEE, the study identifies various impurities that affect battery performance and establishes a trade-off between purification methods and the generation of secondary impurities, ultimately offering a framework for optimizing electrolyte solvent production processes. Journal: *ACS Energy Letters ASAP*
- **A Spatially-Resolved Approach via Quantifying the Dominant Yet Overlooked Orbital-Weighted Energy Contribution for Breaking Solvent Oxidation Chemistry** — score: 0.900 The article introduces a novel Pseudo-Eox approach to assess solvent oxidation stability by focusing on the spatially-resolved energy contributions of the highest-energy electrons rather than relying solely on traditional HOMO-based methods. This method effectively identifies overlooked solvents, such as F3DECF1, and accurately predicts the oxidation stability of various solvents, demonstrating its superiority over conventional ranking models that fail to consider the localized nature of electron loss. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Understanding the Role of Lithium Metal Microstructure in Chemo-Mechanical Instability of Solid-State Batteries** — score: 0.900 This study investigates the impact of lithium metal microstructure on the interface stability of solid-state batteries during discharge and stripping processes. By employing a mesoscale model guided by atomistic simulations, the research highlights how microstructural properties, such as grain size, influence void formation and interfacial contact loss, while also examining the effects of operational factors like temperature and pressure on performance. Journal: *ACS Energy Letters ASAP*
- **Chain-Networked Micro-Silicon Electrodes Inducing Internal Isostatic Compressive Stress for High-Energy and High-Power Solid-State Batteries** — score: 0.800 The article presents a novel Chain-based Networking (CN) architecture for micro-silicon electrodes that enhances the performance of solid-state batteries by inducing internal isostatic compressive stress during lithiation. This innovative design utilizes single-walled carbon nanotubes to create a robust network that improves energy and power densities, achieving remarkable capacity retention after extensive cycling in full-cell configurations, thus overcoming the limitations of traditional silicon anodes in solid-electrolyte-free systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **A Fluidic and Sodiophilic Liquid Metal Current Collector for Dendrite-Free and Stable Anode-Free Sodium Metal Batteries** — score: 0.800 The article presents a novel liquid metal (LM@Cu) current collector that enhances the performance of anode-free sodium metal batteries by reducing the nucleation barrier for sodium deposition, thus preventing dendrite formation. With this improved architecture, the batteries demonstrate remarkable stability, achieving around 80% capacity retention after 100 cycles, surpassing traditional copper-based collectors which degrade within 30 cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Synergistic Ion Regulation and Interface Stabilization Enabled by a PVA/ZIF-8 Hybrid Layer for Dendrite-Free Zinc Anodes** — score: 0.800 The study presents a novel protective interphase (PVA/ZIF-8@Zn) that combines poly(vinyl alcohol) and zeolitic imidazolate framework-8 to enhance the performance of aqueous zinc-ion batteries by mitigating dendrite growth and interfacial reactions. The synergistic properties of this hybrid layer allow for improved zinc ion deposition, exceptional cycling stability, and minimal volumetric fluctuations, resulting in over 2400 hours of stable cycling and substantial capacity retention in full cells. Journal: *Energy & Fuels ASAP*
- **Pre-Agglomeration Granulation for Fluidized Bed Sintering of Geldart Group C LiFePO₄ With Enhanced Electrochemical Performance** — score: 0.800 The study presents a method to transform non-fluidizable Geldart Group C LiFePO₄ powder into fluidizable particles through pre-agglomeration granulation, facilitating efficient fluidized-bed sintering. This innovative approach enhances electrochemical performance by promoting uniform Al³⁺ doping, consistent carbon coating, and improved ion diffusion, resulting in significant capacity retention for lithium-ion batteries while also providing a scalable manufacturing solution for high-performance materials. Journal: *Wiley: Small*

- **Functional-Group Engineering for Durable Solid Electrolyte Interphase Toward Aqueous Copper Batteries** — score: 0.800 This study introduces a functional-group-guided approach to create a robust organic-inorganic hybrid solid electrolyte interphase (SEI) on copper electrodes, significantly enhancing mechanical strength and Cu^{2+} transport efficiency. By using specific additives containing fluorine and sulfur, the resulting SEI allows for improved cycling stability and reversibility in $\text{Cu}||\text{Cu}$ cells and $\text{Cu}-\text{MnO}_2$ full cells, achieving over 2200 cycles while maintaining excellent dimensional stability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Calorimetric and Quasielastic Neutron Scattering Studies on a Mg-Ion Conductor Using a Metal–Organic Framework MIL-101 Involving Acetonitrile** — score: 0.800 The study investigates the high Mg^{2+} conductivity in a metal-organic framework, MIL-101, that incorporates $\text{Mg}(\text{TFSI})$ and acetonitrile, revealing significant insights into the phase behavior and dynamics of acetonitrile molecules within the framework. Through calorimetric and quasielastic neutron scattering experiments, the research identifies a phase transition and a glass transition related to the mobility of acetonitrile, with the presence of $\text{Mg}(\text{TFSI})$ enhancing Mg^{2+} mobility and conductivity due to the coordination of acetonitrile to the ions. Journal: *Journal of the American Chemical Society ASAP*
- **Electrostatically Coupled High-Modulus Interphase with Synergistic Anion–Cation Regulation for Stable Zn Anodes** — score: 0.800 The article discusses the development of a high-modulus interfacial layer designed to enhance the stability of zinc anodes in aqueous zinc batteries by utilizing electrostatic coupling. This innovative approach mitigates issues such as dendrite formation and volume fluctuations during charge cycles, leading to impressive performance metrics, including over 2200 hours of lifespan and more than 2000 cycles of reversible operation, demonstrating its potential for improving the reliability and efficiency of high-performance energy storage systems. Journal: *Journal of the American Chemical Society ASAP*
- **In situ visualization of natural convection-driven concentration stratification in aqueous zinc batteries** — score: 0.800 The article discusses the in situ observation of natural convection-induced concentration stratification in aqueous zinc batteries, highlighting its significance for battery performance. The authors employ advanced visualization techniques to assess how stratification affects the electrochemical processes within the batteries, providing insights into improving efficiency and longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Lamination Pressure Governs the Effectiveness of Sn Buffer Layers in Low-Pressure All-Solid-State Lithium Metal Batteries** — score: 0.800 This study investigates how lamination pressure influences the performance of Sn buffer layers in all-solid-state lithium metal batteries, particularly under low operational pressures. By optimizing lamination pressure, the researchers demonstrated improved interfacial stability and lithium transport in the buffer layer, leading to enhanced battery efficiency and resilience against contact loss, thus highlighting the importance of lamination techniques for practical battery applications. Journal: *ACS Energy Letters ASAP*
- **Boosting Potassium-Ion Conductivity through Heteroatom Effect and Neutral Ligand Coordination for Solid-State Potassium Batteries** — score: 0.800 This study addresses the challenge of finding solid electrolytes with both high ionic conductivity and good interface compatibility for solid-state potassium-ion batteries. Researchers engineered potassium carbolide isomers and their complexes, achieving significant improvements in ionic conductivity through the strategic use of nitrogen heteroatoms and neutral ligands. Notably, the optimized variants exhibited impressive conductivities and stability during prolonged electrochemical cycling, suggesting their potential for practical battery applications. Journal: *ACS Applied Energy Materials ASAP*
- **In Situ Substitution Strategy with High-Valence Cation for Ultra-Stable Aqueous K-Ion Batteries** — score: 0.800 This study introduces a novel strategy to enhance the stability of manganese-based cathodes in aqueous K-ion batteries by using an $\text{Al}(\text{OTF})_3$ substitution in a KOTF electrolyte. This approach effectively minimizes manganese dissolution and Jahn-Teller distortion, resulting in a significant increase in cycling stability and energy density, with impressive retention rates across

thousands of cycles, which could revolutionize energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **MXene-Based Lithium-Ion Anodes: An Outlook on Computational Predictions and Experimental Validation** — score: 0.800 This article discusses the gap between computational predictions of MXene-based lithium-ion anodes and their actual performance in functional cells. The authors argue that this discrepancy stems from the lack of integration between computational models and experimental conditions, which leads to using idealized predictions that do not translate well to real-world applications. They propose strategies such as coordinated theory-experiment studies, machine learning for interatomic potentials, and explicit modeling of electrolytes to better align computational insights with practical anode performance. Journal: *Energy & Fuels ASAP*
- **Ion-Regulated Interfacial Solvation Enables Ether Electrolytes for 4.4 V Lithium Metal Batteries** — score: 0.800 The study presents an innovative ion-regulated approach to optimize interfacial solvation in lithium metal batteries (LMBs), crucial for stabilizing lithium metal anodes and cathodes. By employing K^+ and BH_4^- as ionic regulators, the researchers achieved enhanced electrochemical performance, including consistent Li plating/stripping over 160 cycles and significant capacity retention in full cells, thus advancing the development of ether electrolytes suitable for high-voltage applications. Journal: *ACS Energy Letters ASAP*
- **Dotted Mn₃O₄-Decorated MnO₂ Composites: Fabrication, Interfacial Characteristics, and Multivalent Mn Charge-Storage Behavior** — score: 0.800 This study explores the development of MnO₂/Mn₃O₄ composite electrodes to enhance conductivity, reaction kinetics, and cycling stability for supercapacitor applications. The optimized composite, M2@M4-30, demonstrates outstanding electrochemical performance, with notable capacitance and cycling retention attributed to its unique structure and multivalent manganese states, resulting in improved energy density for practical applications. Journal: *ACS Applied Energy Materials ASAP*
- **Tailoring the Hydrogen-Bond Network via a Universal Additive Approach for Low-Temperature Alkaline Zinc/Ferricyanide Flow Batteries** — score: 0.800 The study presents a novel cryogenic electrolyte for alkaline zinc/ferricyanide flow batteries, combining potassium acetate and urea to disrupt the hydrogen-bond network in water, lowering the freezing point to -18.9 °C while maintaining high ionic conductivity. This optimized electrolyte enables stable battery operation at -20 °C, achieving excellent cycle life and efficiency, thus paving the way for improved performance in low-temperature applications. Journal: *ACS Applied Energy Materials ASAP*
- **Aerobic Dehydrogenation by Coupled Electrochemical Reactions Governs Room-Temperature Proton Defect Chemistry in Functional Oxides** — score: 0.800 The study investigates how the ambient oxygen partial pressure can influence proton defect chemistry in functional oxides at room temperature, specifically through aerobic dehydrogenation in hydrogenated NdNiO₃. By employing a combination of electrochemical and structural analyses, the researchers reveal that coupled anodic proton extraction and cathodic oxygen reduction processes allow for effective modulation of ionic defects, challenging previous assumptions about the limitations of room-temperature adjustments. Journal: *Journal of the American Chemical Society ASAP*
- **Pseudo-Jahn–Teller Effect-Mediated Medium-Spin ($S = 3/2$) Fe for Robust Fe–N Bonds in Acidic Oxygen Reduction and Fuel Cells** — score: 0.700 This study highlights the introduction of a p-block metal (Bi) to iron–nitrogen–carbon (Fe–N–C) catalysts to promote a pseudo-Jahn–Teller distortion that stabilizes Fe in a medium-spin state ($S = 3/2$). This modification enhances the Fe–N bonds, significantly improving catalyst durability under acidic conditions, as demonstrated by greater Fe retention and improved performance metrics in H₂–O₂ proton-exchange membrane fuel cells. Journal: *Journal of the American Chemical Society ASAP*
- **A Hybrid Phosphate–Ether Electrolyte for High-Temperature and High-Safety 18650 Cylindrical Sodium-Ion Batteries** — score: 0.700 The article presents a novel hybrid electrolyte composed of triethyl phosphate and tetrahydrofuran designed for high-temperature sodium-ion batteries, enhancing both safety and performance. This nonflammable electrolyte facilitates improved ion

transport and the formation of durable interphases, resulting in long-lasting capacity retention over numerous charging cycles, while exhibiting no combustion risk under thermal stress. Journal: *ACS Energy Letters* ASAP

- **Physics-Informed Machine Learning for Solid Oxide Cell Materials and Microstructures Under Operando Constraints** — score: 0.600 The article presents a four-level framework for integrating physics into machine learning to enhance the performance of solid oxide cells (SOCs) under dynamic operating conditions. It emphasizes the importance of coupling defect chemistry with real-time operational data, rather than just increasing algorithmic complexity, to enable effective microstructure design and experimental validation. The review critically addresses challenges in extrapolating from static data and proposes a structured approach to ensure that machine learning models produce physically plausible outcomes relevant to SOC performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Suppressing Self-Discharge in Aluminum-Graphite Batteries with a Triple-Function Polymer Electrolyte** — score: 0.600 Aluminum-graphite batteries face significant self-discharge issues that lead to considerable capacity loss. This study introduces a PAN-based polymer electrolyte that acts as an ion conductor, separator, and binder, effectively minimizing self-discharge by reducing anion deintercalation and parasitic reactions. The result is a remarkable capacity retention of over 99% after 24 hours, highlighting the potential for improved long-term stability in energy storage applications. Journal: *ACS Applied Energy Materials* ASAP
- **Reassessing H₂O₂ in Malonic Acid Leaching of Spent-LIB Black Mass: Fenton-Like Ni-Co Oxalate Reprecipitation** — score: 0.600 The study reevaluates the use of hydrogen peroxide in the malonic acid leaching of spent lithium-ion battery (LIB) black mass, revealing that it hinders metal recovery due to oxidative degradation rather than enhancing it. Instead, malonic acid alone effectively recovers high levels of nickel, manganese, and lithium, while also generating oxalate precursors for cathode materials, thus challenging traditional methods and suggesting alternative reducing agents for improved metal recovery. Journal: *ACS Applied Energy Materials* ASAP
- **Modeling Approaches for Polymer-Based Electrolytes in Advanced Batteries: Bridging Processing, Structure, and Performance** — score: 0.600 The article discusses the significance of polymer-based electrolytes in advanced batteries, highlighting their tunable properties and performance variability influenced by processing history. It proposes a modeling framework to bridge the gaps between various scales of behavior and emphasizes the need for integrated workflows that connect formulation, processing, and performance to enhance the understanding and development of these materials in battery applications. Journal: *ACS Energy Letters* ASAP
- **Ethynyl/Thienyl-Functionalized Porphyrin Complexes as High-Performance Cathodes for Electrochemical Energy Storage** — score: 0.600 This study evaluates two porphyrin complexes, CuDET4P and DET4P, as cathodes for lithium organic batteries, highlighting how their ethynyl and thienyl functional groups improve cycling stability and capacity retention. CuDET4P achieves a reversible capacity of 216 mAh g⁻¹ after 100 cycles, demonstrating superior performance due to copper coordination, while both materials maintain high stability at elevated current densities. The findings support the potential of functionalized organic molecules in enhancing energy storage applications. Journal: *ACS Applied Energy Materials* ASAP
- **Interfacial Failure Analysis and LiTaO₃-Mediated Dual-Filler Stabilization of LiTa₂PO₈-Based Composite Solid Electrolytes for Solid-State Lithium Metal Batteries** — score: 0.500 The article discusses the analysis of interfacial failures in LiTa₂PO₈-based composite solid electrolytes and explores how LiTaO₃ enhances stability through a dual-filler approach. This research aims to improve the performance of solid-state lithium metal batteries by addressing critical compatibility issues in the materials used. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Architected Porous Media for Fuel Cells, Electrolyzers and Beyond: A Paradigm for Transport Engineering** — score: 0.500 The article advocates for a “transport-by-design” approach in the development of porous media for electrochemical devices like fuel cells and electrolyzers. It

highlights the potential of advanced manufacturing techniques to create specifically engineered porous structures that enhance transport properties, thereby shifting the focus from traditional material-centric methods to those that prioritize the deliberate optimization of transport characteristics for improved energy conversion and storage performance. Journal: *ACS Applied Energy Materials* ASAP

- **Configurational Entropy and Compositional Effects in Fe-Based Oxide 2D Sheets for Nitrate Electroreduction** — score: 0.400 The article discusses the development of FeCuCoMnZnOx as a multicomponent electrocatalyst for the nitrate-to-ammonia reduction process. It highlights that the catalyst's performance is determined more by the composition and local coordination of the materials than by configurational entropy, with specific attention to how electronic structure and oxidation states influence the reaction efficiency and stability during operation. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **High-Performance Rechargeable Anionic Zn–Cu Batteries with Facile Recyclability** — score: 0.400 The article presents a high-performance rechargeable aqueous zinc-copper battery that addresses significant challenges in energy storage, including cycle life and self-discharge. Utilizing zinc and cuprous bromide complexes as the anolyte and catholyte, this innovative design achieves impressive metrics: a capacity of 4 mAh/cm², over 2700 hours of stable cycling, and excellent energy efficiency, while also enabling nearly complete recycling of materials, contributing to its environmental and economic sustainability. Journal: *Journal of the American Chemical Society* ASAP
- **Infiltrating Metal Adhesive Layer for Mechanically Integrated Dry Electrodes** — score: 0.400 This article presents a novel approach for enhancing the adhesion of dry electrodes in lithium-ion batteries by using a sputter-coated indium adhesive layer that infiltrates the porous structure of graphite films. This method allows for a completely solvent-free manufacturing process, resulting in improved interfacial stability and long-term durability of the electrodes, as demonstrated by high capacity retention and stable cycling performance in battery tests. Journal: *Wiley: Advanced Energy Materials: Table of Contents*